

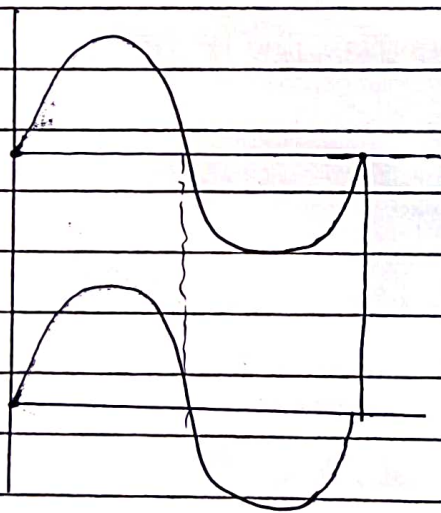
Unit-1 - Wave optics

Interference of light:-

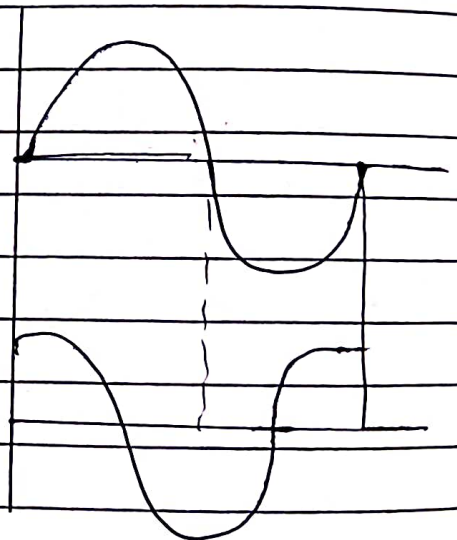
→ Interference of light is the phenomena of redistribution of light energy in a medium on account of superposition of light waves from coherent sources.

⇒ Phase difference:-

→ Two waves that have the same frequencies and different phases are known to have a phase difference and are said to be out of phase with each other.



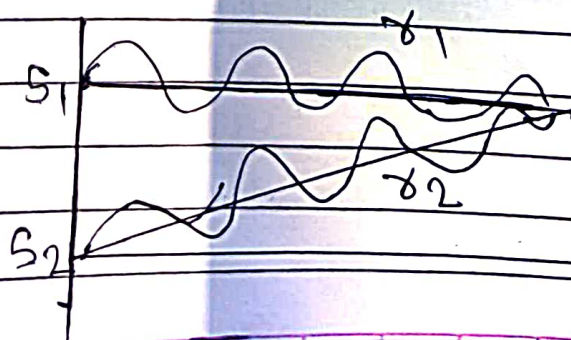
Same phase



out of phase

⇒ Path Difference:-

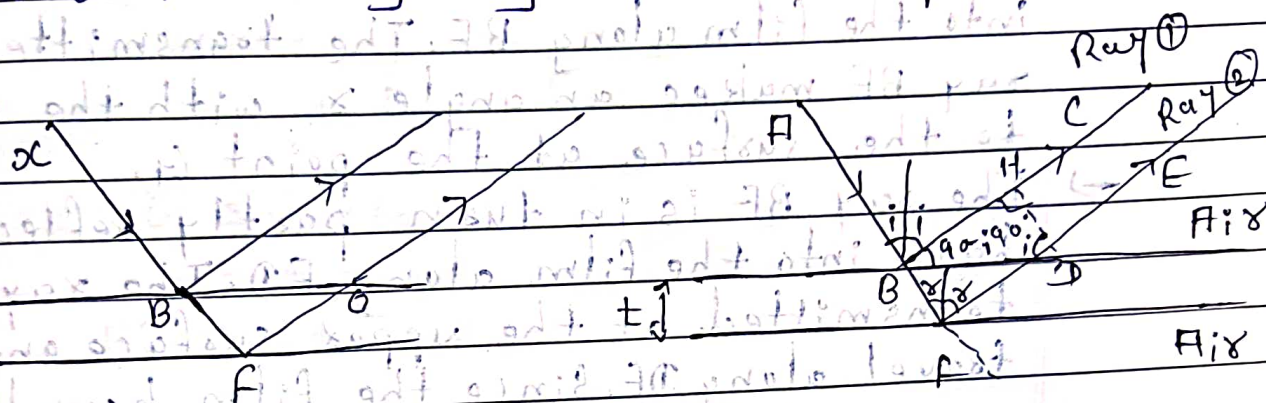
→ Path difference is the difference in the path traversed by the two waves.



$$\delta = x_2 - x_1$$

Interference in thin film:-

→ Thin film interference is a natural phenomena in which light waves reflect back by upper and lower boundaries of a thin film with one another giving interference pattern.



→ When light is incident on a parallel thin film a small portion of it gets reflected from the top surface and a major portion is transmitted into the film.

→ Again a small part of the transmitted component is reflected back into the film by the bottom surface and the rest of it transmitted from the lower surface of the film.

→ Thin films transmit incident light strongly and reflect weakly.

→ After two reflections, the intensities of the reflected rays drop to a negligible strength. Therefore, we consider the first two reflected rays only.

→ These two rays are derived from the same incident ray but appear to come from two sources located below the film.

→ The sources are virtual coherent sources. The reflected wave 1 & 2 travel along parallel paths and interfere.

→ Let us consider plane waves from a monochromatic source falling on the thin film at an angle of incidence i .

→ Part of a ray such as AB is reflected along BC, and part of it is transmitted into the film along BF. The transmitted ray BF makes an angle x with the normal to the surface at the point G.

→ The ray BF is in turn partly reflected back into the film along FD. The ray FD is transmitted at the upper surface and travel along DE. Since the film boundaries are parallel, the reflected rays BC and DE will be parallel to each other.

→ The waves travelling along the paths BC and BFDE are derived from a single incident wave AB. Therefore they are coherent and can produce interference.

Optical path Difference

$$\Delta_0 = \mu(BF + FD) - I(BH) \quad \text{--- (1)}$$

In ΔBFD

$$BF = FD$$

$$\cos x = \frac{FG}{BF}$$

$$\Rightarrow BF = \frac{FG}{\cos x}$$

$$\therefore BF + FD = 2t \quad \text{--- (2)}$$

$$BG = GD$$

$$BD = 2BG$$

$$\tan \theta = \frac{L}{F} = \frac{BG}{FG}$$

$$\Rightarrow BG = FG \tan \theta$$

$$= t \tan \theta$$

$$BD = 2t \tan \theta \quad \text{--- (3)}$$

In $\triangle BHD$

$$\angle HBD = 90^\circ - i$$

$$\angle HBD + \angle BDH = 90^\circ$$

$$90^\circ - i + \angle BDH = 90^\circ$$

$$\angle BDH = 90^\circ - 90^\circ + i$$

$$= i$$

$$\sin i = \frac{L}{K} = \frac{BH}{BD}$$

$$BH = BD \sin i$$

$$BH = 2t \tan \theta \sin i \quad \text{--- (3)}$$

From Snell's law

$$\frac{\sin i}{\sin \theta} = \mu$$

$$\sin i = \mu \sin \theta$$

$$\therefore BH = 2t \tan \gamma (\mu \sin \gamma) \quad \text{--- (3)}$$

$$= 2\mu t \left(\frac{\sin \gamma}{\cos \gamma} \right) (\mu \sin \gamma)$$

$$= \frac{2\mu t \sin^2 \gamma}{\cos \gamma} \quad \text{--- (4)}$$

using eqⁿ (2) & (4) in eqⁿ (1)

$$\Delta a = \mu \left(\frac{2t}{\cos \gamma} \right) \left(\frac{2\mu t \sin^2 \gamma}{\cos \gamma} \right)$$

$$= \frac{2\mu t}{\cos \gamma} [1 - \sin^2 \gamma]$$

$$= \frac{2\mu t \times \cos^2 \gamma}{\cos \gamma}$$

$$= 2\mu t \cos \gamma \quad \text{path difference}$$

→ For constructive Interference (maxima) Bright Fringes

$$2\mu t \cos \gamma = (2m+1) \frac{\lambda}{2}$$

$$m = 0, 1, 2, 3, \dots$$

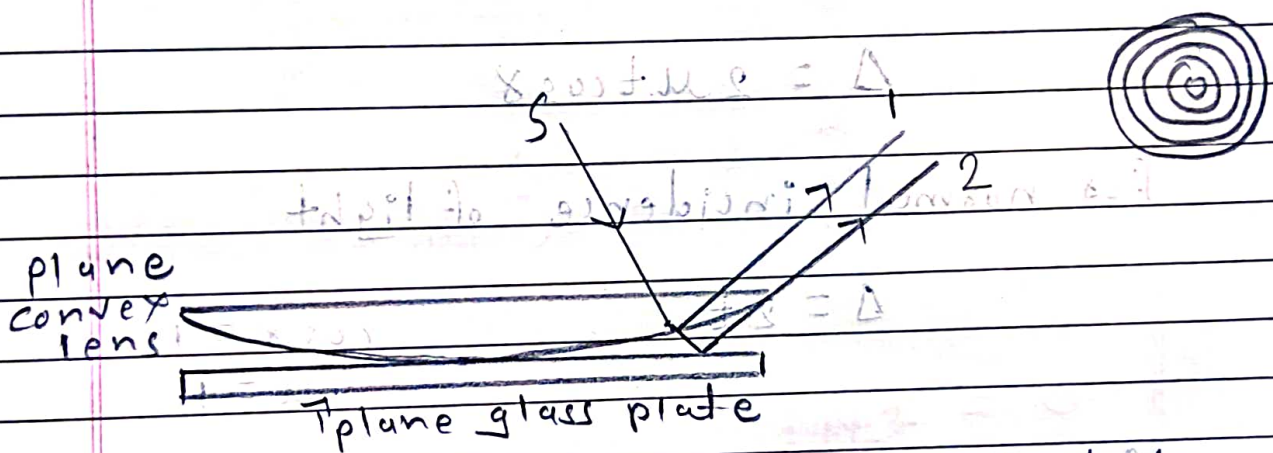
→ For destructive Interference (minima) Dark Fringes

$$2\mu t \cos \gamma = m\lambda$$

$$m = 0, 1, 2, 3, \dots$$

Newton's Ring :-

→ Newton's ring is a phenomenon in which an interference pattern is created by the reflection of light betⁿ two surfaces, typically a spherical surface and an adjacent touching flat surface.



→ Monochromatic light from an extended source is rendered parallel by a lens L. It is incident on a glass plate inclined at 45° to the horizontal and is reflected normally down on to a plane convex lens placed on a flat glass plate.

→ Part of the light incident on the system is reflected from the glass to air boundary. The remainder of the light is transmitted through the air film.

→ It is again reflected from the air to glass boundary. The two rays reflected from the top and bottom of the air film interfere to produce darkness or brightness.

→ The condition of brightness or darkness depends on the path difference betⁿ the two reflected rays which in turn depends upon the thickness of the film.

→ The optical path difference betⁿ the rays is given by

$$\Delta = 2\mu t \cos \theta$$

For normal incidence of light

$$\Delta = 2t$$

$$\cos \theta = 1$$

$$\mu = 1$$

→ For constructive interference (Bright fringes)

$$2t = (2m+1) \frac{\lambda}{2}$$

Maxima

$$m = 0, 1, 2, 3, \dots$$

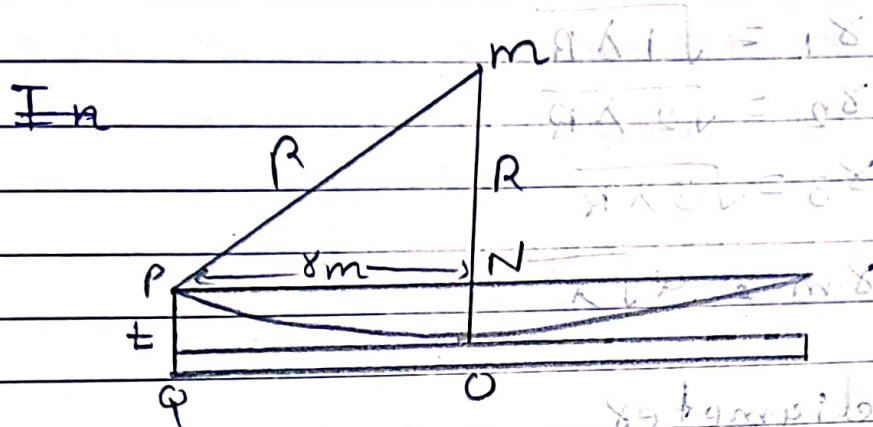
→ For destructive interference (Dark fringes)

$$2t = m\lambda$$

Minima

$$m = 0, 1, 2, 3, \dots$$

- ⇒ Radii of Dark Fringes (Radii of Newton Ring)
 → Let R be the radius of curvature of the lens



In $\triangle PNM$

$$PM^2 = PN^2 + MN^2$$

$$R^2 = x^2 + (R-t)^2$$

$$R^2 = x^2 + R^2 - 2Rt + t^2$$

$$\boxed{x^2 = 2Rt - t^2}$$

As $R \gg t$, $2Rt - t^2 \approx 2Rt$

$$x^2 \approx 2Rt$$

The condition for darkness at ϕ is

$$2t = m\lambda$$

$$t = \frac{m\lambda}{2}$$

$$\therefore x^2 = 2R \cdot \frac{m\lambda}{2}$$

$$= m\lambda R$$

$$\boxed{x = \sqrt{m\lambda R}}$$

→ The radii of dark fringes can be found by using $m = 1, 2, 3, \dots$

$$r_1 = \sqrt{1 \lambda R}$$

$$r_2 = \sqrt{2 \lambda R}$$

$$r_3 = \sqrt{3 \lambda R}$$

$$r_m = \sqrt{m \lambda R}$$

→ Ring diameter

$$D_m = 2 r_m$$

$$= 2 \sqrt{m \lambda R}$$

$$m = 1, 2, 3, 4$$

$$D_1 = 2 \sqrt{1 \lambda R} = 2 \sqrt{\lambda R}$$

$$D_2 = 2 \sqrt{2 \lambda R} = 2 \times 1.414 \sqrt{\lambda R} = 2.828 \sqrt{\lambda R}$$

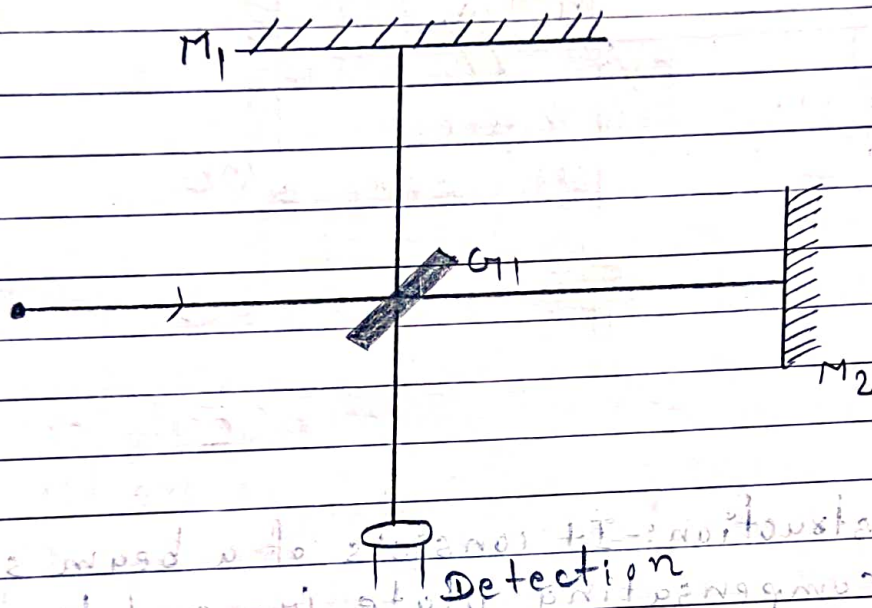
$$D_3 = 2 \sqrt{3 \lambda R} = 2 \times 1.732 \sqrt{\lambda R} = 3.464 \sqrt{\lambda R}$$

$$D_4 = 2 \sqrt{4 \lambda R} = 2 \times 2 \sqrt{\lambda R} = 4 \sqrt{\lambda R}$$

→ The rings get closer and closer as m increases. This is why the rings are not evenly spaced.

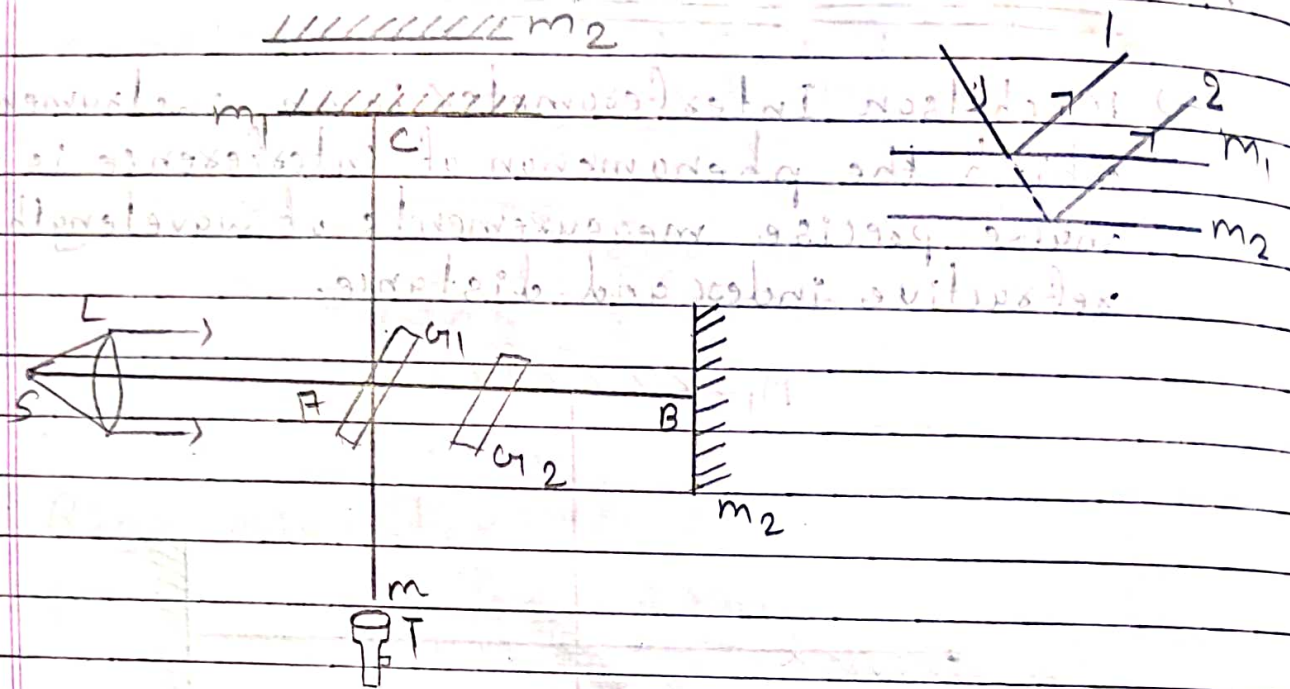
Michelson Interferometer:-

→ Michelson Interferometer is an instrument in which the phenomenon of interference is used to make precise measurements of wavelengths, refractive index and distance.



→ Principle:- In Michelson interferometer, a beam of light from an extended source is divided into two parts of equal intensities by partial reflection and refraction.

→ These beams travel in two mutually perpendicular directions and come together after reflection from plane mirrors. The beams overlap on each other and produce interference fringes.



- Construction:- It consists of a beam splitter G_1 , a compensating plate G_2 and two plane mirrors m_1 and m_2 . The beam splitter G_1 is a partially silvered plane parallel glass plate.
- The compensating plate G_2 is a single plane parallel glass plate having the same thickness as G_1 .
- The two plates G_1 and G_2 are held parallel to each other and are inclined at an angle of 45° with respect to the mirror m .
- The mirror m_1 is mounted on a carriage and can be moved with the help of a micrometer screw. The plane mirrors m_1 & m_2 can be made perfectly perpendicular with the help of fine screw attached to them.
- The interference bands are observed in the field of view of the telescope T.

→ Working :-

- Monochromatic light from an extended source S is rendered parallel by means of a lens L and is made incident on the beam splitter G_1 . It is partly reflected at the back surface of G_1 along AC and partly transmitted along AB .
- The beam AC travels normally towards the plane mirror M_1 and is reflected back along the same path and come out along AT .
- The transmitted beam travels towards the mirror M_2 and is reflected at the back surface of G_1 and comes out along AT .
- The two beams received along AT are produced from a single source hence are coherent. The superposition of these beams leads to interference and produces interference fringes.
- From the fig. it is clearly seen that a light ray starting from the source S and undergoing reflection at the mirror M_1 passes through the glass plate G_1 three times.
- On the other hand in the absence of plate G_2 , the ray reflected at M_2 travels through the glass plate G_1 only once.
- For compensating this path difference a compensating plate G_2 of the same thickness is inserted into the path AB and is held exactly parallel to G_1 .
- If we look into the instrument from T , we see mirror M_1 and in addition we see a vertical image, M_2 of mirror M_2 . Depending on the positions of the mirror, image M_2 may be in front of, or behind or exactly coincident with mirror M_1 .

→ The situation is similar to an air film enclosed between mirrors m_1 and m_2' with the difference that in case of a real film both surfaces, multiple reflections take place, whereas in this case only two reflections take place.

$$m_2' m_1 = d$$

→ path difference

$$D_s = 2d \cos \theta$$

→ For constructive (maxima)

$$2d \cos \theta = (2m+1) \frac{\lambda}{2}$$

$$m = 0, 1, 2, 3, \dots$$

→ For destructive (minima)

$$2d \cos \theta = m\lambda$$